

**Overgeneralization of (Anti)causativization in L2 English Acquisition:
Learners' Acceptability Judgments of Overanticausativized Transitive/Causative
Verbs**

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Abstract

This thesis investigates the overgeneralization of the causative alternation in second language (L2) English, with particular focus on over(anti)causativization errors, whereby non-alternating transitive or causative verbs are accepted in ungrammatical (anti)causative constructions. While previous research has extensively documented overcausativization errors involving unaccusative verbs (e.g., **The magician disappeared the rabbit*), comparatively little attention has been paid to learners' acceptance of non-target anticausative uses such as **The picture painted*.

Building on the manner/result distinction proposed in lexical semantic theory, this study advances a manner/result overgeneralization hypothesis, according to which L2 learners overextend the generalization that only result verbs, which lexicalize the result state of the event, can participate in the causative alternation. Once learners acquire this generalization, they may incorrectly apply it to non-alternating result verbs (*destroy, kill*), while manner verbs are predicted to resist such overgeneralization. This proposal makes predictions that differ from existing accounts based on L1 lexical transfer or the assumption that learners uniformly assign complex event structures to all verbs.

To test this hypothesis, a context-based 7-point acceptability judgment task was administered to 120 Japanese-speaking learners of English across four proficiency levels, as well as a native English control group. The experiment employed a 2×2 design crossing Verb Type (manner vs. result verbs) and Structure (transitive/causative vs. anticausative). All target verbs were non-alternating in both English and Japanese, allowing the effects of lexical semantics to be examined independently of L1 transfer. The judgment data were z-score transformed and analyzed using linear mixed-effects models.

The results revealed that learners at the intermediate proficiency level rated anticausativized result verbs significantly higher than anticausativized manner verbs, while no such asymmetry was observed in lower or higher proficiency groups. This pattern supports the manner/result overgeneralization hypothesis and suggests that intermediate learners possess knowledge of the semantic generalization governing the

causative alternation but overapply it to non-alternating result verbs. The findings indicate that non-target (anti)causativization in L2 English reflects a developmental stage in which learners generalize lexical-semantic principles, rather than mere L1 transfer or undifferentiated event-structure representations.

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Chapter 1: Introduction

The acquisition of verbs has attracted significant interest in the field of both first language (L1) and second language (L2) acquisition. Verb acquisition involves the intricately intertwined morphological, syntactic, and semantic aspects of verbs, posing several challenges for language learners (Becker 2014). In L1 and L2 research, one extensively studied verb class concerns unaccusativity (e.g., Becker & Schaeffer 2013; Kim et al. 2025; Montrul 2005; Perkins et al. 2025; Oshita 2000, 2001; Randall et al. 2004; Shimada & Sano 2007; see also Hirakawa 2024 and Tazaki 2025a for an overview of L1 and L2 studies regarding intransitive verbs). Regarding L2 research, previous cross-linguistic studies have suggested that L2 learners face difficulty with unaccusative verbs due to their structural complexity (Oshita 2001). In particular, non-target-like passivization errors have been central to the investigation in the literature (see, e.g., Chung 2014; Ju 2000; Kim et al. 2025; Oshita 2000, 2001, 2014; Tazaki, 2024b; Tazaki & Hattori 2025; Zobl 1989). Less studied phenomena related to unaccusativity include overcausativization errors, such as **The magician disappeared the rabbit* (Cabrera 2019; Montrul 2001; Zyzik & Bittar 2024). These errors are produced and accepted by L2 learners from various L1 backgrounds (see Hirakawa 1995; Montrul 2001), suggesting the cross-linguistic pervasiveness of overcausativization phenomena in L2 acquisition. What is intriguing about the causativization errors is that such errors are more frequent with unaccusative verbs but less frequent with unergative verbs (e.g., Tazaki 2025b).

This overgeneralization of the causative/unaccusative alternation extends to non-alternating transitive verbs (e.g., *cut*, *paint*). Montrul (2001) found that learners were inaccurate in rejecting non-alternating transitives used in the unaccusative (anticausative) frame (e.g., **The picture painted.*). This finding suggests that learners are likely to struggle with acquiring the causative alternation. Such difficulty is expected, given that the set of verbs that can alternate varies across languages (see Section 2). Due to cross-linguistic variation, learners cannot rely solely on their L1 lexical knowledge or input alone to determine which verbs do not participate in the alternation.

One of the explanations for these overgeneralization errors assumes that these errors stem from the transfer of learners' L1 lexical knowledge. A verb that alternates in their L1 often does not alternate in L2 English and vice versa. For instance, English *happen* can be used only intransitively (*The accident happened.* / **The drunk driver happened the accident.*), while its Japanese counterpart *oki-ru* alternate (*Ziko-ga okotta.* 'The accident happened.' / *Yopparatta doraibaa-ga ziko-o okosita.* 'The drunk driver caused the accident.'). If L1 Japanese learners of English assume that *happen* can be used in the causative construction simply because it can in their language, it is no wonder that learners overuse the causative construction with some intransitive verbs. The same is expected to happen to overanticausativization error.

Another explanation found in the literature is that L2 learners assign a complex event structure template to any verb's lexical semantic representation. This analysis is based on the assumption that only verbs that have a complex event structure (i.e., [[x ACT] CAUSE [y BECOME]]) can participate in the causative alternation (Levin & Rappaport Hovav 1995). According to this analysis, any verbs can alternate between causatives and anticausatives since any verbs can have a complex event structure.

Although the two explanations are experimentally supported, I propose another possible explanation for the over(anti)causativization errors, according to which learners make these errors only with verbs that lexically specify the resulting state or location, what Rappaport Hovav and Levin (2010) and Levin and Rappaport Hovav (2014) call result verbs. The reasoning behind this analysis is that only result verbs, but not manner verbs, can cross-linguistically enter into the causative alternation. If L2 learners "know" the generalization, it is no wonder why they overcausativize unaccusative verbs but rarely unergative verbs (as unaccusative and unergative verbs correspond to result and manner verbs, respectively). Supposing that this analysis is correct, non-target anticausativization is predicted with result transitive verbs (i.e., causative verbs such as *destroy*, *kill*), as opposed to manner transitive verbs (e.g., *touch*, *use*).

Previous research on the acquisition of the causative alternation has primarily focused on L2 learners' overcausativization and overanticausativization errors involving unaccusatives, unergatives, and (non-)alternating causative verbs. However, little is known about learners' knowledge of non-alternating transitive verbs that lexically

encode the manner of action (i.e., manner verbs). To address this gap, the present study employs a context-based acceptability judgment task to investigate whether L2 English learners overgeneralize the causative alternation to non-alternating transitive manner verbs, and to compare their performance with that involving non-alternating causative verbs. Furthermore, by comparing four proficiency groups, the study aims to identify the developmental trajectory in the acquisition of the causative alternation. The result of this experiment suggests that learners are more likely to anticausative result transitive verbs than manner transitive verbs, providing support for the proposed new account.

This thesis is organized as follows. Chapter 2 discusses two types of alternation, focusing primarily on English: what I refer to as the causative alternation and the transitive alternation. Both types involve intransitive and transitive constructions; however, they differ in the grammatical role of the subject in the intransitive variant. In the causative alternation, the subject of the intransitive verb corresponds to the object of the transitive variant, whereas in the transitive alternation, the subject of the intransitive verb corresponds to the subject of the transitive variant. Based on the manner/result distinction, this chapter proposes generalizations for both the causative alternation and the transitive alternation. Chapter 3 reviews two types of overgeneralization errors observed in L2 English learners. One involves causativization errors, in which non-alternating intransitive verbs are incorrectly used in transitive constructions. The other involves anticausativization errors, in which non-alternating transitive verbs are incorrectly used in intransitive constructions. Chapter 4 reviews two existing analyses of these overgeneralization patterns and proposes a novel analysis. Because the proposed analysis yields predictions distinct from those of the existing accounts, an acceptability judgment experiment was designed to adjudicate among them. Chapter 5 presents the experimental design and reports the results. Chapter 6 demonstrates that the proposed analysis is supported by data from intermediate-proficiency learners. This chapter also examines a potential U-shaped developmental pattern in the acquisition of the causative alternation. Although such a pattern was descriptively observed, subsequent statistical analyses did not provide significant support for it. Chapter 7 concludes the thesis.

Chapter 2: Causative Alternation

In English, some verbs can occur in either the Subject-Verb-Object (SVO) construction or the Subject-Verb (SV) construction, while others can appear in only one of the two configurations. For example, *move* can be used in both SVO and SV frames, as shown in (1), but *deliver* can be used only in the SVO frame, as in (2), despite the fact that both verb shares a similar meaning (i.e., *the package* changes the location). Verbs that denote the causation of the change, like (1a), are referred to as causative verbs, and their variants that denote only the change without explicitly expressing its causation, like (1b), are referred to as anticausative verbs.

- (1) a. Kevin moved the package to the office.
b. The package moved to the office.
- (2) a. Kevin delivered the package to the office.
b. *The package delivered to the office.

Interestingly, however, the Japanese counterpart of the causative *deliver*, *todoke-ru*, has an anticausative variant, *todok-u* ‘arrive,’ as exemplified in (3).

- (3) a. Miki-ga pakkeezi-o ofisu-ni todoketa.
Miki-NOM package office-to delivered.
‘Miki delivered the package to the office.’
b. Pakkeezi-ga ofisu-ni todoita.
package-NOM office-to arrived.
‘The package arrived at the office.’

These mismatches between languages can be easily observed. In spite of these mismatches, alternating verbs such as (1) and (3) fall into the same semantic class, namely, change of state/location. Verbs that lexically specify the manner of action, like *sweep* and *wash*, seldom participate in the causative alternation, as will be discussed in

this section. This section will discuss two types of causative alternation, namely, causative alternation and transitive alternation and propose rough generalizations about these alternations.

2.1 Generalization for the Causative/Transitive Alternations

The causative alternation, also known as the causative/inchoative, causative/unaccusative, or causative/anticausative alternation, has long been a central topic of investigation in lexical semantics. Cross-linguistically, verbs that participate in the causative alternation typically denote a change of state, such as *break* and *melt* (Alexiadou 2015; Levin & Rappaport Hovav 1995; Rappaport Hovav 2014). An example involving *break* is shown below:

- (4) a. Brian broke the window.
b. The window broke.

The anticausative variant (4b) describes an eventuality in which the theme, *the window*, undergoes a change of state, while the causative variant (4a) denotes the causation of that change. Since *break* in the anticausative frame like (4b) is an unaccusative verb, the alternation in (4) is sometimes called causative/unaccusative alternation. The nature of the causative alternation remains a topic of debate, and researchers have proposed several analyses without yet reaching a general consensus. Some researchers argue that the anticausative variant is derived from its causative counterpart (e.g., Kageyama 1996; Levin & Rappaport Hovav 1995; Reinhart 2002, 2016), while others assume that the anticausative form is basic and that the causative variant is derived from it (e.g., Rappaport Hovav 2014; see also Levin 2015). This section does not go into the details of derivational approaches, but instead outlines the properties of the causative alternation, drawing on Rappaport Hovav and Levin (2010) and Levin & Rappaport Hovav's (2014) manner/result complementarity.

Non-stative verbs typically lexicalize either manner or result components of verb meaning (Alexiadou 2015; Rappaport Hovav & Levin 1998, 2010): that is, verbs are

classified as either manner or result verbs. Manner verbs lexically specify a manner of carrying out an action, while result verb specify the coming about a result state (see Rappaport Hovav & Levin 2010 for a more detailed definition). This complementarity is explicitly stated by Rappaport Hovav and Levin (2010) and Levin and Rappaport Hovav (2014).

(5) Manner/Result Complementarity:

Manner and result meaning components are in complementary distribution: a verb lexicalizes only one (Levin & Rappaport Hovav 2014: 338).

The manner/result complementarity holds that each verb is associated with only one meaning component—either manner or result. For example, the intransitive verbs *run* and *break* are manner and result verbs, respectively, and can be represented in the event structures in (6a) and (6b).

- (6) a. [x ACT<*RUN*>]
 b. [y BECOME <*BROKEN*>]

The “root” is an idiosyncratic component of verb meaning, represented within angle brackets in (6), while ACT and BECOME are structural predicates that constitute part of the verb’s meaning (Rappaport Hovav & Levin 1998). Manner roots modify the predicate ACT, whereas result roots serve as arguments of BECOME. The causative use of *break* is a result verb that lexicalizes the resultant state (i.e., the broken state) and has a complex event structure consisting of a causing event and the resulting change of state, as represented in (7).

- (7) [[x ACT] CAUSE [y BECOME <*BROKEN*>]]

This manner/result complementarity arises naturally from a lexicalization constraint formulated by Rappaport Hovav and Levin (2010: 25)

(8) The lexicalization constraint:

A root can only be associated with one primitive predicate in an event schema [event structure], as either an argument or a modifier.

In the case of intransitive verbs, the manner/result distinction corresponds to the unergative/unaccusative distinction: unergative verbs lexically specify the manner of action, while unaccusative verbs specify the resulting state or location. That is, unaccusativity is semantically determined but syntactically encoded (Levin & Rappaport Hovav 1992, 1995; see also Burzio 1986; Perlmutter 1978).

Returning to the causative alternation, we arrive at the following generalization:

(9) Only result verbs can enter the causative alternation.

Core verbs participating in the causative alternation are typically change-of-state verbs, which fall under the category of result verbs. This implies that certain subsets of result verbs can participate in the causative alternation, whereas manner verbs cannot (Alexiadou 2015).¹ This generalization, however, does not preclude result verbs such as *kill* and *destroy* from being considered non-alternating, contrary to fact, as illustrated in (10) and (11).

- (10) a. John killed the chickens.
b. *The chickens killed.

- (11) a. Brian destroyed the building.
b. *The building destroyed.

¹ Some manner of motion verbs (e.g., *walk*, *march*) can alternate with an addition of prepositional phrase, as in *The general marched the solders to the tents* (cf. ??*The general marched the solders.*) (Levin & Rappaport Hovav 1995: 111). While these types of sentences have a complex event structure as with lexical causatives, these are different from the typical causative alternation of result verbs (see Levin & Rappaport Hovav 1995).

The verb *destroy* describes an eventuality in which *the building* undergoes a change of state; however, it cannot appear in the anticausative frame. A commonly cited constraint on the unavailability of the anticausative form for such verbs is that verbs whose subjects are restricted to agents or instruments cannot alternate (Levin & Rappaport Hovav 1995; Reinhart 2002, 2016). However, this constraint cannot be maintained, as evidenced by the following examples.

- (12) a. The vandals/the storm/the intense heat destroyed the crops.
 b. The marauders/the poison/the cold killed the chickens.

(Rappaport Hovav 2014: 15)

A comprehensive explanation that accounts for all instances of non-alternating change-of-state transitives has not yet been proposed, but following Reinhart (2002, 2016), I assume that the lexical entries of these verbs “froze” in their transitive form. Which verbs can appear only in the transitive form varies across languages. For example, *destroy* has an anticausative counterpart in Hebrew and Greek (Alexiadou et al. 2006; Reinhart 2016). Therefore, in order for a verb to enter the causative alternation, it must be a result verb, yet being a result verb is not sufficient for a verb to alternate. Verbs that cannot join the alternating group exhibit cross-linguistic variation (see Alexiadou 2015; Reinhart 2002).

In contrast to frozen causative verbs, some anticausative verbs—such as *fall* and *disappear* in English—do not appear in the transitive frame, but similar to non-alternating causatives like *destroy* and *kill*, these verbs are alternating in some languages, such as Japanese (e.g., *otiru* ‘fall’ [anticausative] vs. *otosu* ‘drop’ [causative]).

The causative alternation consists of a pair: a causative variant and an anticausative variant, both denoting the same event, with the latter representing a subevent of the former. The key distinction between the two variants lies in the presence of an external argument: the causative variant includes an external argument expressing the cause, while the anticausative variant lacks one. Although seemingly similar to the causative alternation of result verbs, alternations involving manner verbs

differ both semantically and structurally (Alexiadou & Anagnostopoulou 2013; Levin 2006, 2017; Rappaport Hovav & Levin 1998), as illustrated in (13).

- (13) a. Leslie swept the floor.
b. Leslie swept.

(13a) and (13b) illustrate a manner verb appearing in transitive and intransitive constructions, respectively. In (13b), the object is absent, yet the sentence remains grammatical. I refer to this type of alternation as the transitive alternation, in contrast to the causative alternation. Manner verbs with unexpressed objects, as in (13b), are felicitous when supported by appropriate and explicit contextual cues (Levin 1993: 33; Rappaport Hovav & Levin 1998). The transitive alternation is not available with result verbs, as shown in (14).²

- (14) a. Leslie broke the vase.
b. *Leslie broke.

The two types of transitive verbs are schematized in (15). (In this paper, I refer to manner transitive verbs (non-causative) like (13a) as transitive verbs, and to result transitive verbs (causative) like (14a) as causative verbs.³)

- (15) a. [x ACT<*SWEEP*> y]
b. [[x ACT] CAUSE [y BECOME <*BROKEN*>]]

The transitive event structure in (15a) includes a structural participant *x*, required by the primitive predicate ACT, and a root participant *y*, which is licensed solely by the root.

² Some causative verbs indeed have transitive forms such as *Covid kills*. We do not discuss these verbs since they are stative and irrelevant to our discussion (see Rasia 2025).

³ Causatives have two types: lexical causatives and syntactic causatives (*make*-causative). When I say “causative,” it means a lexical causative in this thesis (see Montrul 2001 for the discussion on the two different causatives).

Since the root participant does not have to be expressed in syntax, the event structure associated with manner verbs allows both transitive and intransitive forms, as seen in (13) (Levin 2006; Rappaport Hovav & Levin 1998). The causative event structure in (15b), on the other hand, consists of two subevents and thus requires two structural participants, namely, *x* and *y*. Since structural participants must be overtly realized in syntax (see Rappaport Hovav & Levin 1998, 2001 for discussion on argument realization), result verbs are naturally excluded from participating in the transitive alternation.

Despite the fact that the mechanisms governing the transitive alternation remain poorly understood, it is widely accepted that, in contrast to result verbs that participate in the causative alternation, manner verbs participate in the transitive alternation (Levin 2006; Rappaport Hovav & Levin 1998). This bifurcation is well accounted for by the manner/result complementarity discussed above. This observation leads to a generalization about the transitive alternation, which stands in contrast to the generalization concerning the causative alternation (9).

(16) Only manner verbs can enter the transitive alternation.

However, this generalization is not sufficient to account for exceptions such as those in (17).

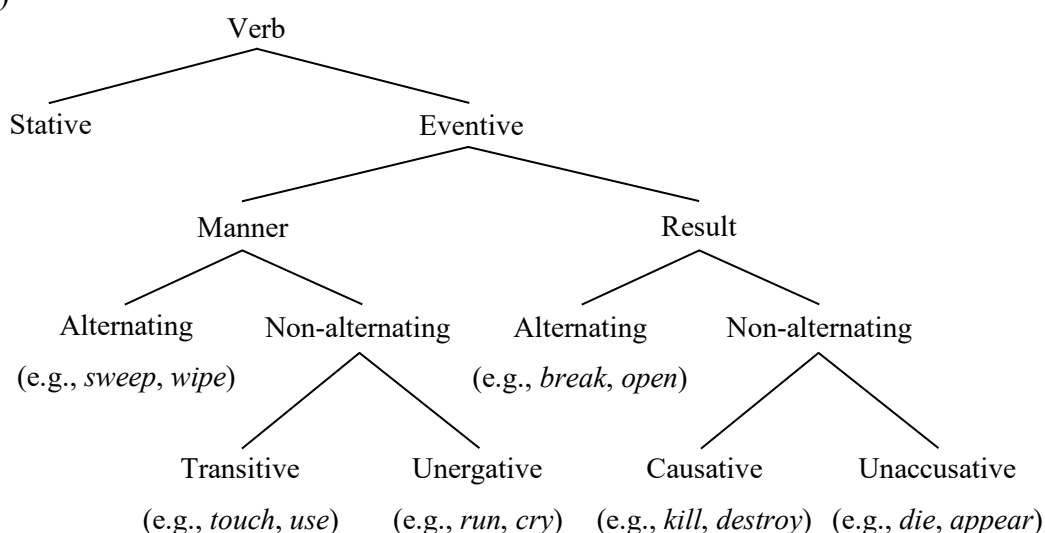
- (17) a. Miki touched something.
b. *Miki touched.

In (17b), the verb *touched* lacks an overt object, resulting in ungrammaticality. Even when the omitted object is contextually recoverable, (17b) remains difficult to accept. Just as certain constraints prevent some result verbs from participating in the causative alternation, some manner verbs are also unable to enter the transitive alternation.

Taken together, eventive verbs fall into either the manner or result category (i.e., they are in complementary distribution). Only manner verbs qualify for participation in the transitive alternation (e.g., *Brian swept the floor* / *Brian swept*), while only result

verbs qualify for the causative alternation (e.g., *Brian broke the vase* / *The vase broke*). Some verbs do not alternate even though they conform to the generalizations for alternating verbs (9) and (16). These verbs vary across languages and must be learned individually through linguistic experience. The classification of verbs is summarized in (18).

(18)



Eventive verbs are divided into manner and result verbs, each of which is further divided into two alternating and non-alternating verbs. Manner verbs and result verbs can enter into transitive alternation and causative alternation, respectively. Non-alternating categories are also subdivided into transitives/causatives and unergatives/unaccusatives.

2.2 Causative Alternation in Japanese

In the preceding subsection, English causative alternation has been discussed. This subsection will focus on the Japanese cases. In English, causatives and anticausative variants look alike; regardless of whether it is used as a causative or an anticausative, the form is identical (e.g., *Kevin opened the door.* / *The door opened.*). Yet, Japanese is a morphologically rich language, typically showing a difference in form between causatives and anticausatives, as already seen in (3). In (3), the causative

todokeru ‘deliver’ and the anticausative *todoku* ‘arrive’ share the same root *todok*. The suffixes (i.e., *-e-* or *-ø-*) indicate which form it is. Japanese has a variety of causative/anticausative suffix combinations. Shibatani (1990) summarizes the basic combination, as shown in Table 1.

Table 1. Transitive/intransitive verb pairs (Shibatani 1990, p. 236)

	Intransitive	Transitive
Group a	<i>-ar</i> (e.g., <i>ag-ar-u</i> ‘rise’)	<i>-e</i> (e.g., <i>ag-e-ru</i> ‘raise’)
Group b	<i>-ø</i> (e.g., <i>ak-u</i> ‘open’)	<i>-e</i> (e.g., <i>ak-e-ru</i> ‘open’)
Group c	<i>-e</i> (e.g., <i>ar-e-ru</i> ‘be ruined’)	<i>-as</i> (e.g., <i>ar-as-u</i> ‘ruin’)
Group d	<i>-ø</i> (e.g., <i>wak-u</i> ‘boil’)	<i>-as</i> (e.g., <i>wak-as-u</i> ‘boil’)
Group e	<i>-e</i> (e.g., <i>sak-e-ru</i> ‘split’)	<i>-ø</i> (e.g., <i>sak-u</i> ‘split’)

As can be seen from Table 1, most alternating verbs have different suffixes between a causative and an anticausative.⁴

A crucial question relevant here is whether the generalization of the causative alternation, as stated in (9) holds in Japanese as well. The generalization predicts only result verbs to be qualified as alternating verbs, and it is indeed borne out.

(19) Result verb *break*

- a. Miki-ga kabin’-o watta.
 Miki-NOM vase-ACC broke
 ‘Miki broke the vase.’
- b. Kabin’-ga wareta.
 vase-ACC broke
 ‘The vase broke.’

(20) Manner verb *punch*

⁴ One exception is *masu* ‘increase.’ It does not change form regardless of whether it is a causative or an anticausative.

- a. Miki-ga karesi-o nagutta.
Miki-NOM boyfriend punch
'Miki punched her boyfriend.'
- b. *Karesi-NOM nagureta.
boyfriend-NOM punch
'The boyfriend was punched.'

Other alternating verbs are also listed below.

- (21) *akeru* / *aku* 'open', *sizumeru* / *sizumu* 'sink', *tokasu* / *tokeru* 'melt', *kowasu* / *kowareru* 'break'

These alternating verbs listed above are all result verbs since they lexically specify the result state. This indicates that the generalization (9) holds. Like English cases, some result verbs do not alternate even though they are categorized as result verbs.

learners are sensitive to the semantic distinction between the two classes of intransitive verbs, and that they tend to overgeneralize the generalization stated in (9).

In addition to these observations from production data, several experimental studies involving acceptability judgment tasks have reported similar findings (e.g., Cabrera & Zubizarreta 2005; Hirakawa 1995, 2000; Matsunaga 2005; Oshita 1997; Tazaki, 2024a). For example, Tazaki (2025b) conducted a 7-point acceptability judgment task with 30 Japanese-speaking learners of English whose TOEIC scores ranged from 550 to 745 (corresponding to CEFR level B2). The task included four unaccusatives and four unergatives in the non-target-like lexical causative construction, as shown in (23).⁵

- (23) a. *Bruce fell the expensive cup from the table.⁶ (unaccusative)
b. *Mary danced her friend yesterday. (unergative)

The results showed that unaccusatives were more likely to be accepted in the causative frame than unergatives (mean scores: 3.59 vs. 2.39; $p = .0097$), further suggesting learners' sensitivity to the unaccusative/unergative distinction. A noteworthy finding in Tazaki (2025b) is that the mean ratings for lexically causativized unaccusatives and unergatives were both below 4, indicating that even intermediate-level learners are somewhat aware of the unacceptability of causative uses of non-alternating unaccusatives as well as unergatives. This leads to the conclusion that some underlying linguistic principle is at work in L2 acquisition.

Thus far, we have reviewed overcausativization errors (e.g., (22)). We now turn to overanticausativization errors, in which non-alternating causative verbs (e.g., *kill*, *cut*) are used in intransitive NP-V constructions, with the subject NP denoting an entity undergoing a change of state or location. Compared to causativization errors,

⁵ In his study, unaccusatives and unergatives were presented in the non-target passive structure as well (see Tazaki 2025b for a detailed information on the study).

⁶ The unaccusative verb *fall* has a causative counterpart, *fell*, which is now primarily used to mean 'cut down trees.' However, because this causative use is highly contextually restricted, *fall* is treated as a non-alternating verb.

anticausativization errors have received less attention. One influential study in this area is Montrul (2001). To investigate the overgeneralization of the causative alternation, Montrul designed a picture judgment task that included non-alternating causative verbs in four constructions: transitive, *make*-causative, intransitive, and *get*-passive. (The task also included alternating causatives, unaccusatives, and unergatives; however, the present discussion focuses primarily on non-alternating causatives.) Each sentence was accompanied by a matching picture. A total of 18 Turkish-speaking and 29 Spanish-speaking L2 English learners were asked to judge whether each target sentence was natural in the context depicted in the picture, using a scale from -3 (completely unnatural) to 3 (completely natural). The Turkish-speaking learners were classified as low-intermediate in proficiency, while the Spanish-speaking learners were of intermediate proficiency. The results suggest that the Turkey group failed to reject ungrammatical intransitives (mean = -0.29), while the Spanish group succeeded in rejecting them (mean = -1.75), similarly to native speakers of English (mean = -2.42).⁷ The acceptability difference between the two L1 groups is difficult to interpret, as the groups differ in both proficiency and L1 background. One possible explanation is that the difference arises from L1 transfer. Another possibility is that lower-proficiency learners (the Turkish group) possess incomplete knowledge of the causative alternation, whereas higher-proficiency learners (the Spanish group) have acquired this knowledge (see Montrul 2001 for a detailed discussion). Supporting evidence for the latter explanation comes from a similar study conducted by Cabrera (2019) with 60 Spanish-speaking learners of English. In her picture judgment task, non-alternating causatives, non-alternating unaccusatives, unergatives, and alternating causatives were tested in the anticausative frame. Crucially, participants were divided into three proficiency groups. Of these, only the lowest-proficiency group failed to reject non-alternating causatives. This finding suggests that Spanish learners, like the Turkish learners in Montrul's study, tend to overanticausativize non-alternating causatives. These results support the analysis

⁷ Her study also showed that L1 Turkey learners failed to reject unaccusatives (0.07) and unergatives (-0.42) in the causative construction. This finding is different from other previous studies. I do not have clear answer to the difference (see Cabrera 2019 for some methodological problems of Montrul (2001)). I leave the reason for this result open for further investigation.

that learners initially overextend the causative alternation to non-alternating verbs. What is particularly important here is that L2 English learners have difficulty judging not only ungrammatical causatives but also ungrammatical anticausatives.

In summary, L2 English learners tend to allow non-alternating unaccusatives—but not unergatives—to appear in causative constructions involving both an external and internal argument. They also permit non-alternating transitives to appear in intransitive constructions, which have only an internal argument. However, no study to date has examined overanticausativization errors involving transitive verbs (i.e., manner verbs). If transitive verbs are found to resist such overgeneralization, this would provide evidence that result verbs are eligible to enter the causative alternation, while manner verbs are not.

Chapter 4: Analyses of Overgeneralization of the Causative Alternation

Section 3 reviewed overgeneralization errors involving non-alternating verbs. Previous studies have identified two key findings: (i) unaccusative verbs (i.e., result verbs), but not unergative verbs (i.e., manner verbs), are prone to overcausativization; and (ii) non-alternating causatives (i.e., result verbs) are often overanticausativized. This section reviews two previously proposed analyses (Sections 4.1 and 4.2) and introduces a novel analysis (Section 4.3). These three accounts generate distinct, testable predictions regarding anticausativization errors involving non-alternating verbs (see Section 5.3).

4.1 Two Existing Accounts for Causative Alternation Errors

4.1.1 The Influence of L1 Verbs on L2 English Verbs

One of the fundamental differences between first and second language acquisition is that L2 acquisition is influenced by learners' L1 knowledge. Because of the critical differences, the relation between L1 and L2 must be considered in L2 acquisition. Specifically, when the L2 grammar permits broader usage than the L1 grammar, learners can acquire the relevant structures through positive evidence alone. In contrast, when the L2 grammar is more restricted, learners must rely on some form of negative evidence (Inagaki 2001; Nagano 2015). For example, the range of alternating verbs available in Japanese and English demonstrates a superset–subset relationship. In English, disappearance verbs such as *disappear* and *vanish* cannot be used in the causative construction (see 24b), while in Japanese, these verbs can undergo alternation (see 25b).

- (24) a. The coin disappeared.
b. *The magician disappeared the coin.

- (25) a. Koin-ga kieta.

coin-NOM disappeared

‘The coin disappeared.’

- b. Mazisyan-ga koin-o kesita.

Magician-NOM coin-ACC disappeared

‘*The magician disappeared the coin.’

This contrast suggests that Japanese permits a broader range of verb classes to alternate than English. Based on such observations, Nagano (2015) argues that Japanese-speaking learners of English are prone to overcausativization errors with intransitive verbs that fall in a lexical semantic class that undergoes alternation in Japanese. In Nagano’s study, Japanese learners of English evaluated non-alternating unaccusatives that alternate in Japanese and exhibited some inaccuracy in their judgments of causative uses of these verbs. Although Nagano’s study supports the L1 transfer account of causative errors, it did not include verb classes that fail to alternate in both L1 Japanese and L2 English. Therefore, we cannot conclusively determine from his study whether L1 lexical entries are being transferred to the L2.

Recently, Tazaki (in preparation) tested this hypothesis by examining overcausativization errors committed by Japanese-speaking learners of English at the intermediate proficiency level (range of participants’ TOEIC scores between 550 and 745). He designed an acceptability judgment task including three types of intransitive verbs in the causative construction. One type is unergatives that do not alternate in both L2 English and participants’ L1 Japanese, Ir Is unaccusatives that do not alternate in both English and Japanese, and the other is unaccusatives that do not alternate in English but do alternate in Japanese. Tazaki predicted that if negative transfer of Japanese lexicon to English occurred, Japanese learners were expected to accept Japanese alternating/English non-alternating unaccusatives in a causative construction but reject Japanese/English non-alternating intransitives—unaccusatives and unergatives. This prediction was borne out by his experiment. Learners successfully rejected causativized unergatives and unaccusatives, both of which do not alternate in Japanese, but were less accurate at rejecting causativized unaccusatives that alternate in Japanese. Although his study supports L1 transfer analysis, his study, however, included

only intermediate learners, which does not clarify whether this analysis applies to more beginner learners.

Summarizing, the L1 transfer analysis analyzes causativization errors as negative transfer from L1. This analysis received support from Tazaki (in preparation). What has not yet been examined is whether other proficiency learners also suffer from causative errors. Tazaki's study included only intermediate learners, veiling the developmental trajectory. In addition to overcausativization phenomena, this analysis makes a falsifiable prediction in terms of anticausativization errors: Causative verbs that cannot appear in the anticausative construction in the learners' L1 are not expected to be accepted in the anticausative construction in L2 (see also Section 5.6).

4.1.2 Overgeneralization of Complex Lexical-Semantic Representation

Departing from L1-based accounts, Montrul (2001) proposes that in L2 acquisition, learners incorrectly overgeneralize the complex lexical-semantic event structure assigned to causative verbs to verbs that are typically associated with simpler event structures, such as unaccusatives and unergatives. In other words, in the interlanguage grammar of L2 learners, unaccusatives and unergatives that do not alternate in causative-anticausative constructions in target-like English are nonetheless assigned a complex event structure consisting of two subevents: a causing event and a caused event. This lexical-semantic misanalysis enables learners to treat non-alternating verbs as if they could undergo the causative alternation, since verbs with a complex event structure are generally able to alternate (Levin & Rappaport Hovav 1995). This overgeneralization reflects an initial "default dyadic template" hypothesis, according to which learners incorrectly assume that all verbs can undergo the causative/inchoative alternation. Although native-like sensitivity to verb-specific syntactic constraints may develop at higher proficiency levels, early L2 grammars lack clear differentiation among verb classes, resulting in persistent transitivity errors.

Montrul's (2001) analysis accounts well for her findings, which show that low-intermediate English learners struggled to reject lexically causativized unaccusatives and unergatives, as well as anticausativized non-alternating causatives. The participants

appeared to assign a complex event structure—consisting of a causing and a caused subevent—to unaccusative and unergative verbs, and to apply anticausativization to non-alternating causatives. However, her default dyadic template analysis has received limited support from subsequent studies, as learners’ causative errors have been shown to involve primarily unaccusatives (see Section 3). If, as the analysis suggests, learners initially assign a complex event structure to all verbs—regardless of their transitivity—the differing patterns between unaccusatives and unergatives remain puzzling.

Despite several findings that challenge Montrul’s hypothesis, the scarcity of studies examining the developmental trajectory of causative alternation in L2 acquisition makes it difficult to dismiss the hypothesis entirely. One possibility is that learners initially assign complex event structures to all verbs, as Montrul proposes. However, once they come to distinguish between semantic verb classes—those that specify a causing event and those that specify a resulting state—they begin to overgeneralize the causative alternation primarily to unaccusative verbs.⁸

4.2 Proposal: Overgeneralization of Causative Alternation

Any account of the (anti)causative errors must account for two key observations: (i) overcausativization errors are more frequent with unaccusatives than with unergatives, and (ii) overanticausativization errors occur with non-alternating causatives. The root of unaccusatives typically serves as the argument of a BECOME predicate in the event structure, while that of unergatives as the modifier of an ACT predicate (Levin & Rappaport Hovav 1998). As discussed in Section 2, many unaccusative verbs—such as *break* and *open*—but not unergatives, participate in the

⁸ If this line of explanation turned out to be true, we would expect to observe a relative U-shape development, instead of an absolute U-shape (see Oshita 2014). At the initial state, learners’ grammar subsumes lexical semantic structure of all verbs under a complex one, and (anti)causativize all verbs. At the second stage, they begin to realize the manner/result distinction, (anti)causativizing only result verbs but not manner verbs. At the final stage, they fully acquire the idiosyncratic lexical entries, expunging (anti)causativization errors. See Section 6.2 for relevant discussion. I, however, do not adapt this analysis. See Section 6.

causative alternation. This supports the generalization that only verbs that lexically specify a result state can enter the causative alternation, as stated in (9) and repeated below in (26).

(26) Only result verbs can enter the causative alternation. (= (9))

If learners are sensitive to the manner/result complementarity, they may infer generalization (26) and overapply it to non-alternating unaccusative verbs. Unergative verbs, by contrast, are less likely to be affected by this overgeneralization, as manner verbs do not satisfy generalization (26). Given that any L2 learners show some sensitivity to semantic and syntactic contrast between unaccusatives and unergatives in previous investigations (e.g., Hirakawa 2001; Oshita 2000; Yuan 1999, among many others), it is safe to say that learners are aware of the manner/result complementarity.

If this line of analysis is correct, anticausativization errors are to be expected, since non-alternating causatives typically satisfy the generalization (26). Learners may treat causative verbs such as *cut* and *paint* as alternating and accept them in anticausative constructions.

This novel proposal makes new predictions that differ from those of Montrul's default dyadic template hypothesis. It predicts that learners will accept non-alternating causatives, but not non-alternating transitives, in anticausative constructions. (27b) is expected to be acceptable to L2 learners, whereas (28b) is not.

- (27) a. The construction workers destroyed the building.
b. *The building destroyed.
c. [[x ACT] CAUSE [y BECOME <DESTROYED>]]

- (28) a. The baby touched the apple.
b. *The apple touched.
c. [x ACT<TOUCH> y]

This asymmetry is predicted because the verbs in (27) lexically encode a result state (see 27c), whereas those in (28) encode a manner of action (see 28c). Only the former qualifies for the causative alternation, and only the latter for the transitive alternation (see generalization (16)). Recall that structural participants must be overtly realized in syntax, whereas root participants may be unexpressed. Hence, structural participants *x* and *y* in (27c) must be overt; in contrast, the root participant *y* in (28c) does not have to be overt, though the structural participant *x* must be expressed. The predicted asymmetry between causative and transitive verbs in susceptibility to (anti)causativization errors is not expected by Montrul's analysis, which assumes that both verb types can be assigned a complex event structure and thus alternate. This prediction remains untested and calls for empirical investigation to evaluate the validity of the manner/result overgeneralization hypothesis.

Because some verbs meet the generalization (26) yet do not alternate, learners must acquire this idiosyncratic usage individually. They cannot rely on their L1 to determine which verbs do not undergo the causative alternation, as these verbs vary cross-linguistically (Alexiadou et al. 2006; Reinhart 2002, 2016). Furthermore, positive evidence is unavailable (Nagano 2015). As a result of these challenges, learners are expected to exhibit (anti)causativization errors over an extended period of acquisition.

4.3 Chapter Summary

This section has discussed two existing accounts and proposed a novel account for the over(anti)causativization errors committed by a wide variety of L1 groups. The first one assumes that L2 learners produce non-target (anti)causativizes based on their L1 lexical knowledge (Section 4.1). If their L1 allows a given causative verb to be used in the anticausative frame, it is likely to be used as an anticausative in L2 language even if the L2 does not allow such usage. The second one attributes causative error to learners' overgeneralization of complex event structure (i.e., [[*x* ACT] BECOME [*y* BECOME]]) to non-alternating intransitive verbs that have simple event structure (i.e., [*x* ACT] or [*y* BECOME]). Given that anticausatives are derived from complex event structure, anticausativization errors are also expected (Section 4.2). Section 4.3

proposed a new hypothesis, referred to as the manner/result overgeneralization hypothesis, according to which, once learners become aware that only result verbs can participate in the causative alternation, they overapply this generalization to non-alternating result verbs

Chapter 5: Experiment

Most previous studies have focused on causativization errors (e.g., **The magician disappeared the rabbit.*), while anticausativization errors (e.g., **The wall painted black.*) have received comparatively less attention. This study employs an acceptability judgment task to investigate these understudied overgeneralization errors and to test the predictions proposed by the novel analysis (see Section 4.3).

The research aim is twofold: (i) to examine if L2 English learners anticausativize causatives (i.e., result verbs) more frequently than transitives (i.e., manner verbs) and (ii) to examine if there are any differences in learners' behavior to causatives/transitives between proficiency groups. These two goals correspond to our two research questions provided below.

RQ1: Do L2 English learners show higher acceptability for non-alternating causative verbs than non-alternating transitive verbs when both are used in the non-target anticausative construction?

RQ2: Are there any differences between proficiency groups in L2 English learners' acceptability judgment on non-target anticausative constructions?

Furthermore, the findings of this study are expected to provide new insight into L2 learners' implicit knowledge of manner/result verbs and the learnability of this manner/result distinction.

5.1 Participants

For the present experiment, 120 Japanese-speaking learners of L2 English were recruited via the Japanese crowdsourcing service *Lancers* (<https://hosting.lancers.jp/lp/client/>).⁹ Participants received 130 yen as compensation for

⁹ *Lancers* was chosen as a platform among other Japanese crowdsourcing service because previous experimental studies have used it (e.g., Asami 2024; Tazaki 2025b, c, in preparation). Additionally, Tazaki (2025b, in preparation) conducted an acceptability judgment task using *Lancers* to collect

their participation. On the crowdsourcing service, I asked only Japanese native speakers who have taken the TOEIC test within the past year. Despite the requirements, because there was no mechanism to restrict participation to qualified individuals only, some unqualified participants may have completed the experimental task. Participants who did not provide their TOEIC scores or whose scores were obtained more than one year before the time of this experiment were excluded from the analysis. By the exclusion process, eight of them were screened out; that is, 112 participants remained. The participants' TOEIC scores ranged between 400 and 965 (mean = 641.2). Their age ranged between 20 and 62. 82 of them were male, 38 were female, and one did not answer. Additionally, 27 participants were excluded since they were considered to lack knowledge of the causative alternation (see Section 5.2 for details). The remaining 85 participants were divided into four proficiency groups based on their self-reported TOEIC scores. Those whose scores are between 400 and 545 were classified as lower proficiency (Low group), those whose scores are between 550 and 645 as intermediate proficiency (L-Intermediate group), those whose scores are between 650 and 745 as high-intermediate proficiency (H-Intermediate group), and those whose scores are between 750 and 965 as high proficiency (High group). The range of TOEIC scores for the High group is wider than that of the other groups because the number of participants whose scores are high was smaller. 15 participants were classified as the Lower group, 27 as the L-Middle group, 26 as the H-Middle group, and 17 as the Higher group. In addition to the L2 learners, 15 native speakers of English were also recruited via *Prolific* as a control group.

Since crowdsourcing has been less employed to recruit participants in L2 research (especially studies that conduct acceptability judgment tasks), this section briefly discusses the advantages and disadvantages of using crowdsourcing services for L2 studies. Let us start with the disadvantages. One big concern of using crowdsourcing can be the reliability of the data. Unlike face-to-face traditional experiments, we cannot keep participants under observation while they are performing the task. This might

data for L2 research and showed similar results to previous studies that were conducted without using crowdsourcing service.

increase the risk of them cheating. This risk also applies to studies that are conducted online. Even if participants are taking the task with friends or with a dictionary, researchers have no choice to exclude these unwelcome participants. Another potential problem concerns compensation. When a crowdsourcing service is used, researchers must pay money to participants who complete the task. Because of the monetary motivation of participating in the study, they might tell a lie to be an eligible subject. For example, in this study, only native Japanese speakers who have a TOEIC score are qualified as participants. If one is not a native speaker of Japanese or has not taken the TOEIC, they might provide false information, such as saying that they are a native Japanese speaker or have taken the TOEIC. We cannot determine who is lying, possibly including inappropriate data for analysis. Furthermore, it can also be problematic that we have difficulty in controlling the range of ages of participants. In the current study, the age range is from 20 to 62, which is even wider than many studies in the literature (in many studies, participants are undergraduate students). Because of this variety in terms of ages, our results are potentially influenced by this factor (Hummel 2021; Saville-Troike & Barto 2017)

Despite these drawbacks, there are several advantages as well. For one thing, recruiting participants is extremely easy (Nakatani 2024). A typical judgment experiment requires at least a couple of weeks to recruit participants before experimenting. In contrast, by using a crowdsourcing service, we can collect data within about a day. The data from 120 participants of this study were collected around a day after I posted the recruitment announcement on *Lancers*. Another advantage is that participants do not get nervous because they can take the task in a relaxing situation. In the face-to-face experiment style, the experimenters are typically in the same room where participants are taking the task. These situations may impose stress on participants. In an online setting, on the other hand, participants are free from the stress of being kept under observation.

Taken together, the use of crowdsourcing services for recruiting participants and administering an experiment has both advantages and disadvantages. The situation is also true for traditional recruitment and experiments. In this study, one of our main objectives is to examine the trajectory of acquisition of causative alternation, which

requires collecting a relatively large amount of data. In this case, we can take advantage of a crowdsourcing service. This is a main reason for the use of crowdsourcing in this study (for discussion about conducting online experiments, see Ishihara & Hamada 2025).

5.2 Materials

To investigate L2 learners' knowledge of the causative alternation, an acceptability judgment task was designed. This experiment had a 2 x 2 design crossing VERB TYPE (manner verb vs. result verb) and STRUCTURE (causative/transitive frame vs. anticausative frame). Because the central research question concerns whether learners overgeneralize the causative alternation to non-alternating causative/transitive verbs, two verb classes were selected: manner transitive verbs and result transitive verbs. Five verbs were selected from each class. In (29), the Japanese counterparts of each English verb are indicated in parentheses next to the corresponding English verb.

- (29) a. Manner verb: *touch* (*sawaru*), *kick* (*keru*), *use* (*tukau*), *purchase* (*kau/koonyuu-suru*), *push* (*osu*)
- b. Result verb: *destroy* (*hakai-suru*), *damage* (*dameezi-o-ataeru*), *remove* (*haizyo-suru*), *create* (*tukuru/sakusee-suru*), *construct* (*kensetu-suru*)^{10,11}

¹⁰ The verbs *destroy* and *damage* can be translated as *kowasu* and *kizutsukeru*, respectively, both of which have anticausative counterparts (*kowareru* and *kizutsuku*). This constitutes a potential confounding factor that may have influenced the results of the experiment. This limitation was identified only after the experiment had been conducted. I am grateful to Hiroshi Aoyagi (p.c.) for drawing my attention to this possibility.

¹¹ The result verbs used in this study are classified into three lexical-semantic subcategories. *Destroy* and *damage* are classified as verbs of change of state, *remove* as a verb of removal/disappearance, and *create* and *construct* as verbs of creation. Differences among these subtypes within the same verb class (i.e., result verbs) may have influenced the results of the experiment. I am grateful to Hiroshi Aoyagi (p.c.) for drawing my attention to this potential issue.

All causative verbs are non-alternating in both English and Japanese, ensuring that Japanese-speaking participants cannot overanticausativize these verbs based on L1 knowledge. Two lexicalizations of each verb were constructed for each of the two conditions—the grammatical transitive/causative frame as in (30a) and the ungrammatical anticausative frame as in (30b).

- (30) a. NP₁-V-NP₂: e.g., *The angry elephant destroyed the yellow fence.*
 b. *NP₂-V: e.g., **The yellow fence destroyed last night.*

The 40 target sentences were divided into two lists using a Latin Square design, resulting in 20 targets in each list. All target sentences consist of animate subjects and inanimate objects, so that participants do not interpret the syntactic subject in the anticausative sentence as an agent of the event denoted by the target verb. Each sentence was presented with a preceding contextual sentence written in the participants' native language, Japanese (for the test for the control L1 English group, they were written in English). As anticausative constructions are typically used when the cause of an event is unclear or unspecified (see Rappaport Hovav 2014), explicit agents of the event denoted by the target verbs were omitted in contextual sentences for the target verbs appearing in the anticausative frame. Moreover, since anticausative sentences do not have a direct object, contrary to the causative counterpart, time adverbs such as *two days ago* were attached to the anticausative sentences. Contextual sentences also help participants to interpret the target sentence as the researcher expects (Sorace 1996). Examples of the critical items are listed in (31), where sentences to be judged are italicized. For all experimental sentences, refer to the Appendix. An example of the screen participants view is also presented below in Figure 1.

- (31) a. NP₁-V_{MANNER}-NP₂
 The famous scientist bought a special camera to observe a certain phenomenon. And then...
The famous scientist used the special camera.
 b. NP₂-V_{MANNER}

In the university laboratory, there was a special camera that no one was allowed to use. However...

The special camera used yesterday.

c. NP₁-V_{RESULT}-NP₂

The young chef found spoiled milk in the refrigerator of the restaurant. And then...

The young chef removed the spoiled milk.

d. NP₂-V_{RESULT}

Since last week, spoiled milk had been in the refrigerator of the Italian restaurant. However...

The spoiled milk removed today.

The screenshot shows a user interface for a linguistic experiment. At the top, there is a progress bar with a blue segment on the left and the word "progress" below it. Below the progress bar, a sentence is displayed in both Japanese and English: "大学の駐車場にレンガの壁が設置されることが計画された。そして..." and "The brick wall created two days ago." Below the sentence, there is a rating scale consisting of seven numbered boxes (1 to 7) and the words "間違っている" (wrong) on the left and "正しい" (correct) on the right. At the bottom, a instruction in Japanese says "数字をクリックするか、数字キーを押してください。" (Click the number or press the number key).

Figure 1. Example of participants' screen (*create* in the anticausative frame)

In addition to the 20 target sentences, 40 fillers were mixed, and their order was pseudo-randomized. Thus, each participant judged 60 sentences. The number of acceptable and unacceptable sentences was counterbalanced. As for 20 target stimuli in one list, 10 were grammatical and 10 ungrammatical. As for fillers, 20 were grammatical and 20 ungrammatical. The initial four sentences to be rated in the task were fillers, so that they serve as a practice.

Additionally, three causative alternation verbs (*break*, *open*, *sink*) were included in fillers. The three verbs were presented twice, once in the causative frame and once in

the anticausative frame, both of which are acceptable sentences in English. These verbs serve as a pretest to see whether participants have knowledge of causative alternation. It would be meaningless to examine the overgeneralization if participants lack basic knowledge of causative alternation. Data from only those who judged four out of the six sentences as “correct” were analyzed (see Section 5.4).

5.3 Procedure

The task was a 7-point scale acceptability judgment task (Goodall 2021). The experiment was conducted online using PCIBex (Zehr & Schwarz 2018). Each participant was instructed to read a contextual sentence and to rate the correctness of the following sentence on a scale of 1–7 (1 = incorrect, 7 = correct). There was no time limitation (see Ionin 2021 for discussion about timed/untimed judgment task). Participants were not allowed to go back to the sentence they had already judged to change their answer.

The experiment proceeds as follows. A participant read the explanation and the instructions of this study. Only those who understand this experiment and agree to participate proceed. On the next page, they were asked questions about their TOEIC scores and their history of learning English. Then, the judgment task starts. The first four sentences were practice sentences (participants do not know that these four sentences were practice), so that participants can familiarize the scale and the task before they get to any experimental items (Goodall 2021). After the four, the actual experiment begins. Participants can take part in this experiment online whenever and wherever they want.

5.4 Data Analysis

Before analyzing the data obtained in this experiment, a screening procedure was implemented. As briefly mentioned in Section 5.2, participants who did not show their knowledge of causative alternation were excluded. Because of the scale of correctness (1–7), participants who assign scores of 1, 2, 3, or 4 to more than two

grammatical causative and anticausative sentences—that is, those who fail to accept correct causative and anticausative sentences—were excluded from the analysis. As mentioned in Section 5.1, 27 participants were excluded because they failed to judge four or more grammatical causative and anticausative sentences.¹²

The judgments provided by the participants were transformed into z-scores prior to analysis to correct for potential scale biases among participants (Schütze & Sprouse 2013). The standardized ratings were analyzed separately for each group with linear mixed-effects models, with VERB TYPE (manner verb vs. result verb) and STRUCTURE (NP₁-V-NP₂ vs. NP₂-V) as fixed factors and subjects and items as random intercepts ($Z_Score \sim Verb_Type * Structure + (1 | Participant) + (1 | Verb)$). When a significant interaction was detected, post hoc pairwise comparisons using estimated marginal means (emmeans; Lenth 2024) were conducted to clarify the nature of the interaction. Furthermore, to examine the differences in acceptability between different proficiency groups, ratings of each condition were separately analyzed using linear mixed-effects models ($Z_Score \sim Verb_Type * Structure * Proficiency + (1 | Participant) + (1 | Verb)$). Pairwise comparisons were conducted using *emmeans* package with Tukey adjustments for multiple comparisons ($model, pairwise \sim Proficiency | Verb_Type + Structure, adjust = "tukey"$). In particular, to examine proficiency-related differences within each condition, we compared proficiency levels separately for each combination of VERB TYPE and STRUCTURE. These analyses were performed in the statistical software *RStudio* (RStudio Team 2022) using the *lme4* package (Bates et al. 2015).

¹² The number of learners who failed to rate correct (anti)causative sentences was relatively high. The high number might be due to learners' reluctance to accept grammatical anticausative sentences. In Montrul (2001), L1 Turkish and L1 Spanish learners of L2 English rated causative alternation verbs significantly lower in the anticausative construction than in the causative construction (see Montrul 2001 for details). Since the number of the grammatical (anti)causative sentences is small in the present study, no detailed discussion takes place, but one possible reason why anticausative sentences are rated lower may be due to English learners' dispreference for Subject-Verb word order with an internal argument in the subject position (i.e., unaccusative construction) (see Oshita 2001, 2002, 2004, 2014 for relevant discussions).

5.5 Methodological Justification

In the second language research, there are several means to investigate learners' linguistic knowledge (see Ellis 1991; White 2023; see also Schütze & Sprouse 2013, for a comprehensive summary of judgment experiments). This subsection discusses three types of judgment tasks widely used in L2 research—forced-choice, yes/no (binary choice), and Likert scale tasks—and illustrates why the Likert scale task is employed for the present study.

A forced-choice task requires participants to choose one option out of two (or more) that they feel is the most acceptable. For example, participants are presented with a causative sentence (*The girl damaged the toy.*) and an anticausative sentence (*The toy damaged.*) at the same time and asked to choose the more acceptable one. One of the advantages of this task is that it allows for a direct comparison between the two conditions, as it presents two critical conditions simultaneously. Because of the nature of the task, it increases statistical power (Schütze & Sprouse 2013). This task, however, is not suitable for the current study for some reasons. Firstly, forced-choice tasks are not ideal for detecting the degree of (un)acceptability. As previous studies have shown, L2 learners are generally sensitive to the acceptability of (non-)target-like (anti)causative (see, e.g., Montrul 2001; Tazaki 2024a, b, c, 2025b, in preparation; see also Section 3). Given these results, even if the manner/result overgeneralization hypothesis is correct, result verbs in the anticausative construction are not expected to be judged as more acceptable than their grammatical counterparts in the causative construction. In these cases, a forced-choice task might not be able to detect the subtle difference in acceptability between manner and result verbs. Secondly, even if participants equally accept or reject the two sentences on the same page, this task forces them to choose one, which can lead to an incorrect conclusion.

A yes-no task is also commonly employed in L2 research (Ishihara & Hamada 2025). In this task, participants are instructed to judge one sentence as acceptable or not; that is, there are two choices for each sentence, acceptable/yes or unacceptable/no. Unlike a forced-choice task, a yes-no task does not force participants to choose which

sentence is more acceptable, addressing the problem of a forced-choice task. However, like a forced-choice task, it is not sensitive to small differences between conditions.

A Likert scale task is what this study employs. In this task, participants rate sentences on a numerical scale, for example, in this study, ranging from 1 to 7. This task addresses the issues of the other two tasks. It does not ask participants to choose a more acceptable sentence and is sensitive to small differences. For these reasons, the Likert scale task is the most appropriate judgment task among the three discussed here (other than the three, a magnitude estimation task is also sometimes used in L2 research (Sorace 1996)).

Although a Likert scale task is the most suitable judgment task, this task has some limitations, one of which concerns the use of the numerical scale (see also Schütze & Sprouse 2013; Goodall 2021). One might, for example, treat the interval between 1 and 2 as bigger than that between 3 and 4. To minimize this issue, following Goodall (2021, p. 17), this study designed the acceptability task in the following ways. First, the numerals were spread out evenly on the scale; that is, the physical interval between numbers is identical (see Figure 1). Second, the only edges of the scale were labeled (i.e., 1 labeled as “incorrect” and 7 as “correct”), and no labels were placed on the intermediate points on the scale. Third, all the numerals look alike. Fourth, the scale consists of an odd number of numerals, which avoids a situation where participants consciously or unconsciously divide them into “correct” and “incorrect” ones.

5.6 Predictions

The proposed hypothesis is based on the distinction between manner and result verbs (Levin & Rappaport Hovav 2014; Rappaport Hovav & Levin 2010). It claims that learners acquire generalizations (9/26) and (16) and subsequently overextend them to non-alternating verbs. According to this view, learners assume that result verbs may undergo the causative alternation, while manner verbs may not. Thus, this account predicts overanticausativization for result verbs (i.e., causative verbs), but not for manner verbs (i.e., transitive verbs). In other words, the interaction of Verb Type and Structure should be observed.

This predicted pattern is not expected by the other two existing accounts (see Sections 4.1 and 4.2). The L1 lexical transfer analysis argues that learners only overanticausativize causative verbs that can be used in the anticausative frame in their L1, implying that when a given causative verb is non-alternating in their L1, it will not be accepted in the anticausative construction. Simply put, the analysis predicts that Japanese-speaking participants similarly judge the acceptability of both causative and transitive verbs in this experiment, as all the transitive verbs, regardless of the verb types—manner or result verbs—are non-alternating in their L1 Japanese as well. The prediction of the default dyadic template hypothesis is the same, though the reasoning behind it is different. The hypothesis argues that any verb can be assigned a complex event structure, meaning that both causative and transitive verbs can have a complex event structure. If it is correct, learners should judge both causatives and transitives similarly.

5.7 Results

Let us first start with descriptive statistics. A summary of mean scores and standard deviations was displayed in Table 2. Native English speakers showed expected results, accepting causatives and transitives in the causative/transitive construction and rejecting these in the anticausative construction. This indicates that the task functions as expected. L2 English learners also showed a similar pattern to native speakers, successfully accepting causative/transitive sentences and rejecting non-target anticausative sentences. In spite of their relatively successful judgment, a close look at the mean ratings of each condition reveals an interesting tendency. Across all proficiencies, L2 learners tend to judge causative result verbs as more acceptable than transitive manner verbs in the anticausative construction. This propensity supports the manner/results overgeneralization hypothesis.

Table 2. Mean row scores and standard deviations

Group	Structure	Mean (SD)
Native control (<i>n</i> = 15)	NP ₁ -V _{MANNER} -NP ₂	6.53 (1.00)
	*NP ₂ -V _{MANNER}	1.32 (0.77)
	NP ₁ -V _{RESULT} -NP ₂	6.43 (1.10)
	*NP ₂ -V _{RESULT}	1.44 (0.83)
Low (<i>n</i> = 15)	NP ₁ -V _{MANNER} -NP ₂	5.60 (1.23)
	*NP ₂ -V _{MANNER}	3.33 (1.75)
	NP ₁ -V _{RESULT} -NP ₂	5.43 (1.41)
	*NP ₂ -V _{RESULT}	3.68 (1.86)
Intermediate (<i>n</i> = 27)	NP ₁ -V _{MANNER} -NP ₂	6.35 (1.04)
	*NP ₂ -V _{MANNER}	2.56 (1.20)
	NP ₁ -V _{RESULT} -NP ₂	6.22 (1.20)
	*NP ₂ -V _{RESULT}	3.07 (2.18)
H-Intermediate (<i>n</i> = 26)	NP ₁ -V _{MANNER} -NP ₂	5.67 (1.74)
	*NP ₂ -V _{MANNER}	2.75 (2.07)
	NP ₁ -V _{RESULT} -NP ₂	5.62 (1.68)
	*NP ₂ -V _{RESULT}	3.00 (2.20)
High (<i>n</i> = 17)	NP ₁ -V _{MANNER} -NP ₂	5.81 (1.58)
	*NP ₂ -V _{MANNER}	3.74 (1.95)
	NP ₁ -V _{RESULT} -NP ₂	5.82 (1.48)
	*NP ₂ -V _{RESULT}	4.15 (2.05)

Figures 2, 3, 4, 5, and 6 exhibit mean z-scores and error bars of native controls and L2 learners by groups. A series of linear mixed-effects models was fitted to the z-scored acceptability ratings. The design included two within-subject factors, VERB TYPE (Manner vs. Result) and STRUCTURE (causative vs. anticausative), crossed with participant group (four learner groups and a native control group). Across all groups, there was a consistent and robust effect of STRUCTURE: sentences in the causative frame were judged as significantly more acceptable than those in the anticausative frame (native

control: $\beta = -1.901$, $SE = 0.052$, $t = -36.803$, $p < .001$; Low: $\beta = -1.055$, $SE = 0.141$, $t = -7.500$, $p < .001$; Intermediate: $\beta = -1.501$, $SE = 0.081$, $t = -18.453$, $p < .001$; H-Intermediate: $\beta = -1.241$, $SE = 0.097$, $t = -12.777$, $p < .001$; High: $\beta = -1.023$, $SE = 0.128$, $t = -7.978$, $p < .001$). Regarding VERB TYPE, there were no differences between manner and result verbs (native control: $\beta = -0.045$, $SE = 0.052$, $t = -0.867$, $p = .387$; Low: $\beta = -0.097$, $SE = 0.146$, $t = -0.669$, $p = .509$; Intermediate: $\beta = -0.018$, $SE = 0.093$, $t = -0.195$, $p = .847$; H-Intermediate: $\beta = -0.04$, $SE = 0.097$, $t = -0.042$, $p = .966$; High: $\beta = 0.022$, $SE = 0.148$, $t = 0.151$, $p = .882$). The interaction was also not significant in native control, Low, H-intermediate, and High groups (native control: $\beta = 0.087$, $SE = 0.073$, $t = 1.186$, $p = .236$; Low: $\beta = 0.175$, $SE = 0.199$, $t = -0.879$, $p < .380$; H-Intermediate: $\beta = 0.117$, $SE = 0.137$, $t = 0.853$, $p = .394$; High: $\beta = 0.148$, $SE = 0.181$, $t = 0.815$, $p = .416$). The Intermediate group, however, showed an interaction effect ($\beta = 0.258$, $SE = 0.115$, $t = 2.243$, $p = .025$). A post hoc test was performed on the data from the Intermediate group. The test revealed that result verbs were rated higher than manner verbs in the anticausative frame (estimate = -0.230 , $SE = 0.095$, $t = -2.591$, $p = .017$).

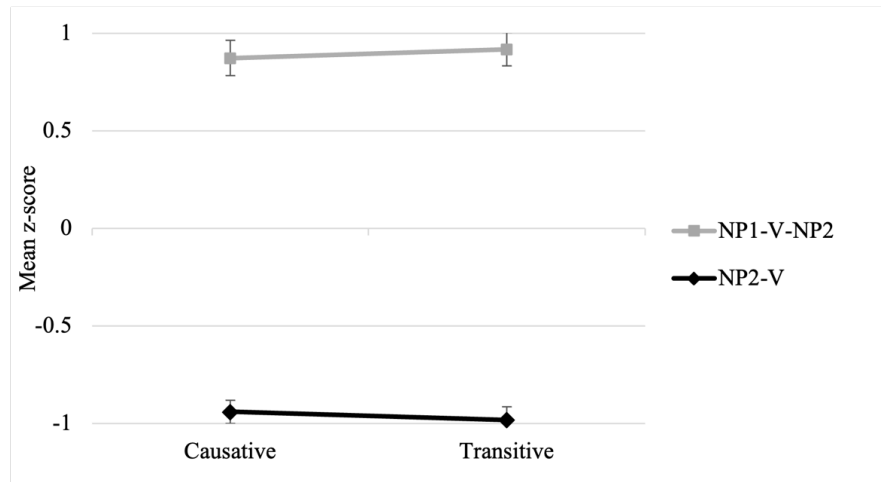


Figure 2. Z-scores and error bars (native English speaker)

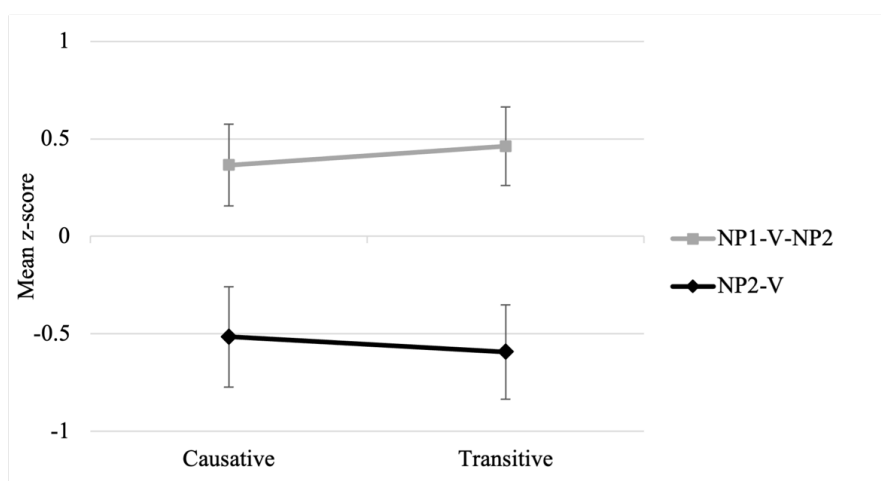


Figure 3. Z-scores and error bars (Low proficiency)

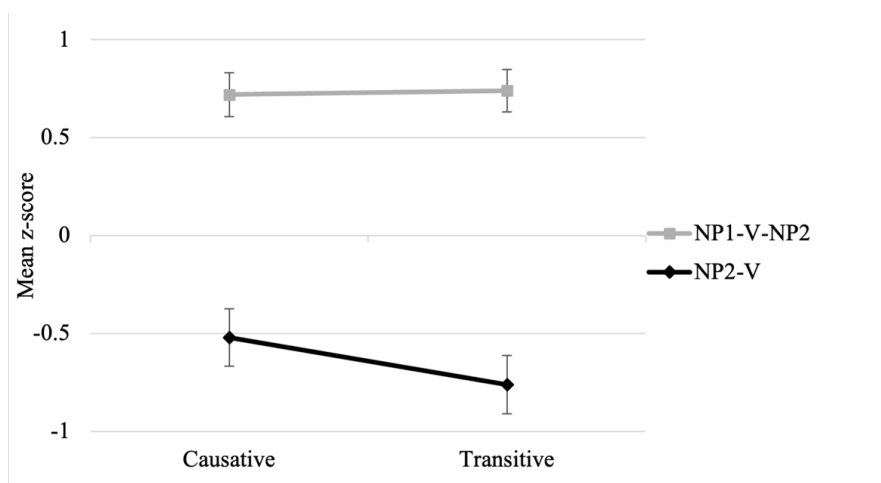


Figure 4. Z-scores and error bars (Intermediate proficiency)

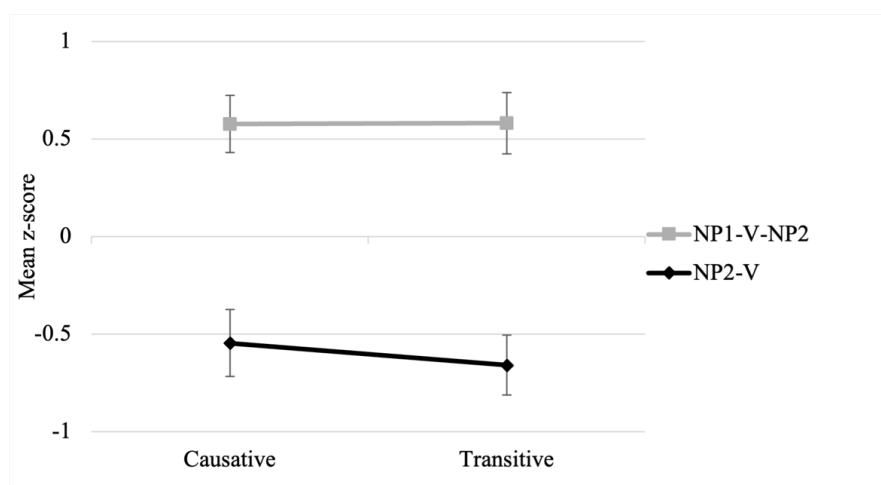


Figure 5. Z-scores and error bars (H-Intermediate proficiency)

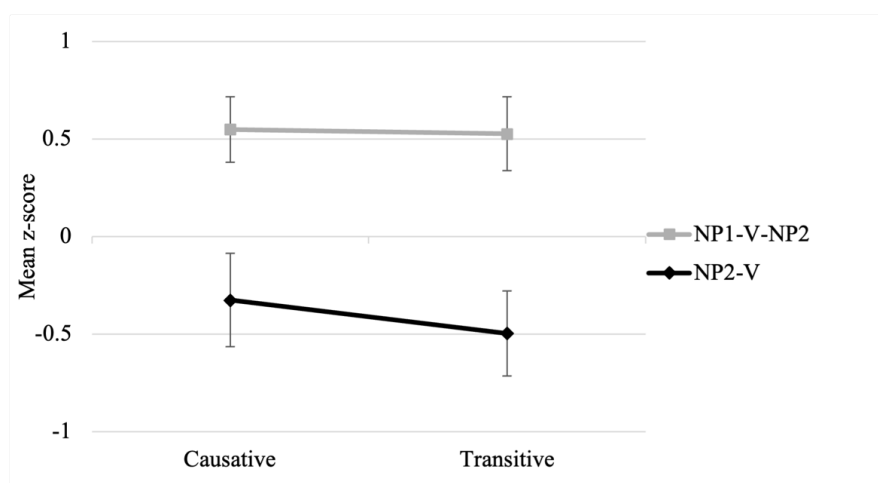


Figure 6. Z-scores and error bars (High proficiency)

The results indicate that L2 English learners at the intermediate proficiency level overgeneralize the causative alternation to non-alternating causative result verbs, but not to non-alternating transitive manner verbs. This is expected by the manner/result overgeneralization hypothesis, according to which learners overapply the generalization that only result verbs can alternate to non-alternating result verbs. This study also suggests that learners start to rate both manner and result verbs in the anticausative construction similarly.

Regarding the acceptability scores on each condition, the results showed an interesting development: the Intermediate group showed the most successful judgments. For convenience, this developmental difference is displayed in Figure 7.

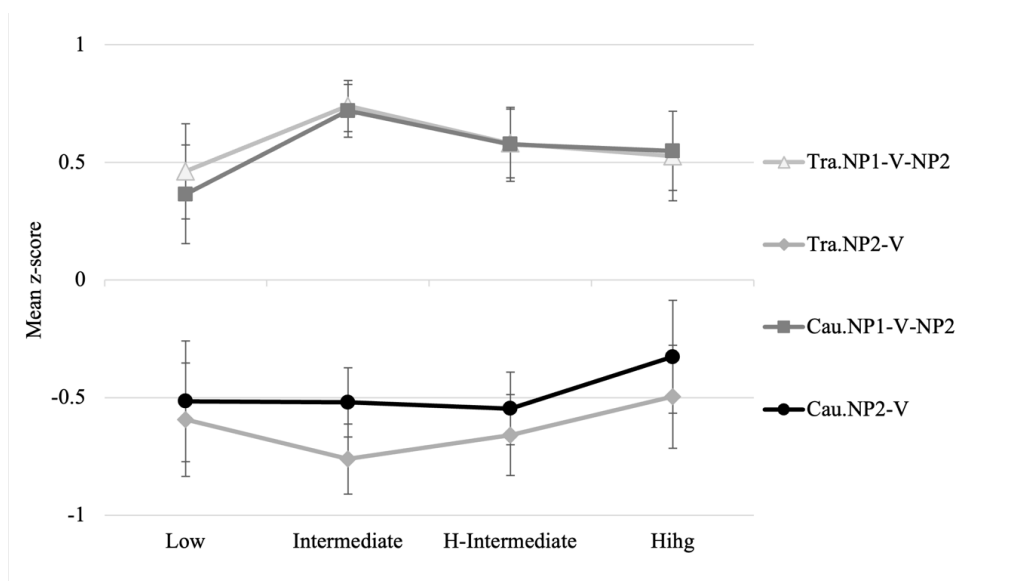


Figure 7. Z-scores and error bars (all groups)

Statistical significances were not found between proficiency groups in ratings of each condition ($p > .05$), except for the differences between the Low and the Intermediate groups in the scores on causative verbs in the causative construction ($p = .018$).

Chapter 6: Discussion

The primary goal of this thesis is to reveal the cause behind non-target (anti)causativization observed in the L2 English interlanguage grammar. Specifically, this study has focused on anticausativization errors. For convenience, the two research questions are repeated below.

RQ1: Do L2 English learners show higher acceptability for non-alternating causative verbs than non-alternating transitive verbs when both are used in the non-target anticausative construction?

RQ2: Are there any differences between proficiency groups in L2 English learners' acceptability judgment on non-target anticausative constructions?

This study conducted an acceptability judgment experiment with 120 L2 English learners to answer the two questions. Concerning the first research question, the answer is affirmative. The Intermediate groups rejected overanticausativized transitive verbs (i.e., manner verbs) more successfully than overanticausativized causative verbs (i.e., result verbs). Regarding the second research question, the answer is also yes. The judgment scores obtained from the Low, H-Intermediate, and High groups did not show any interaction between verb type (transitive vs. causative) and structure (NP₁-V-NP₂ vs. NP₂-V), while the only Intermediate group did. Crucially, since all the experimental verbs are non-alternating, even in participants' native language, Japanese (but see footnote 10), we cannot attribute this behavior of the Intermediate group to L1 transfer (see Section 3.1).

These results, especially those from the Intermediate group, suggest that (i) L2 English learners are sensitive to the distinction between manner and result verbs and (ii) know the generalization that result verbs, but not manner verbs, can participate in the causative alternation. These results support the manner/result overgeneralization hypothesis. These findings also indicate that learners are quite accurate in accepting transitive/causative verbs in the NP₁-V-NP₂ structure and rejecting the NP₂-V structure from the beginning. In this study, even the lowest proficiency learners successfully

judged (in)correct sentences, showing the ease of acquiring the correct knowledge of causative alternation.

6.1 Transition from the First to the Second Stage

Now, a new question arises: Why only the Intermediate group showed a more successful judgment on transitive verbs than causative verbs in the anticausative construction? In other words, why did the Low group not show any difference between the two types of verbs? To think about the possible answer to the question, it is important to pay close attention to the difference between the Low and Intermediate groups. See Figure 7 again. Among the two conditions—transitive and causative verbs in the anticausative construction, the judgment on the only transitive one got better from the Low groups to the Intermediate group (though the difference was not statistically significant), with the judgment on the causative one remaining, more or less, the same.

There are at least three possibilities to explain the results of the Low group: (i) The Low learners are insensitive to the manner/result distinction, (ii) the Low learners use their L1 lexicon, and (iii) the Low learners are insensitive to the generalization for the causative alternation (9/26) repeated as in (32).

(32) Only result verbs can enter the causative alternation.

The first possibility is compatible with Montrul's (2001) default dyadic template hypothesis (see Section 3.2), according to which any verbs can have a complex event structure template at the lexico-semantic representation. Under this possibility, in the interlanguage grammar of the Low learners, transitive verbs (manner verbs) are also assigned a complex event structure like causative verbs (result verbs). Because both types of verbs have a similar complex event structure, the ratings of the two types were also identical. This analysis, however, appears to me extremely difficult to hold, if not impossible. For illustration, take one example from the current study. (32) is an example. The target sentence has a transitive verb *purchase*.

(33) Context: A white car displayed at the dealership had not been sold at all for five years. However...

Target: *The white car purchased today.

(34) a. *[[x ACT_{<PURCHASE>} car] CAUSE [car BECOME]]

↓ $\Leftarrow$ anticausativization

b. *[car BECOME <??>]

If this analysis is correct, the Low learners interpreted the sentence *the white car purchased* as *the white car* undergoing some change, as schematized in (34b). However, from the context, it is clear that *the white car* does not undergo any change in state or location. For this reason, the first possibility is untenable.

The second possibility assumes that the Low learners rely on their L1 knowledge (Hirakawa 2000; Montrul 2001; Nagano 2015; Tazaki in preparation; see also Section 3.1). Since all the experimental verbs are non-alternating in Japanese as well, if the learners assume that non-alternating causative verbs in their language are also non-alternating in L2 English, their similar judgment on transitive and causative verbs is explainable.¹³

The third possibility is that the learners do not know the generalization (32). If they have no idea what types of verbs alternate, it is also expected that transitive and causative verbs are similarly judged in the anticausative frame.

The last two possibilities are both compatible with the results of the Low group. This time, let us think about these possibilities from a developmental perspective; that is, how is the transition from the initial stage to the next stage achieved, where learners judge anticausativized causative verbs as more acceptable than anticausativized transitive verbs? As this study shows that learners at the intermediate proficiency level perform better at rejecting anticausativized transitive verbs, it is crucial to explain the developmental change. Let us begin with the second possibility that the Low group used

¹³ Note that some of the result verbs used in this study can be interpreted as alternating verbs in Japanese (see footnote 10).

L1 knowledge. This analysis can account for the results of the Low groups but fails to do so for the results of the Intermediate group because it does not predict the Intermediate group's better performance on anticausativized transitive verbs in comparison with anticausativized causative verbs. Under this analysis, since both transitive and causative verbs used in this experiment are non-alternating verbs in participants' L1, even the Intermediate group is expected to judge both of them similarly, contrary to the results. These reasons reject this L1 transfer analysis as a primary explanation of the Low and Intermediate groups' judgment of anticausativized transitives and causatives.

The third possibility assumes that the Low learners are not sensitive to the generalization of the causative alternation (32), but the Intermediate learners are. If this analysis is correct, the Low learners distinguish between manner and result verbs but do not know which type can enter the causative alternation. The Intermediate learners become aware of the generalization that only result verbs can alternate and judge result verbs (causative verbs) as slightly more acceptable in the non-target anticausative frame than manner verbs (transitive verbs). Incidentally, our results show that they correctly reject both types of verbs (their mean judgment scores of both types are below the middle point, 4), which suggests that they know that verbs used in this experiment are unacceptable in the anticausative construction but feel like anticausativized result verbs are more acceptable than anticausativized manner verbs due to the generalization (32).¹⁴

¹⁴ A question that arises here is why even low proficiency learners know that verbs used in this study are non-alternating transitive/causative verbs. I do not have a decisive answer to the question but raise some possible explanations. One is that learners initially use L1 lexical knowledge so that they do not accept transitive/causative verbs that are not alternating in their L1 (Hirakawa 2000; Montrul 2001; Tazaki in preparation). This was rejected above because of its failure to explain the differences between the results of the Low and the Intermediate groups. However, a possible scenario is that at the early stage, learners are insensitive to the generalization of the causative alternation and use L1 knowledge. Another potential explanation is that learners do not accept verb usage that does not exist in the input. As all experimental verbs are unacceptable in the anticausative frame in native English speakers, learners are highly unlikely to be exposed to the verbs in the anticausative frame. Since learners have never had exposure to these ungrammatical structures, they do not accept overanticausativized transitive/causative verbs.

Now, the third possibility appears to give unified account for the differences between the Low and the Intermediate groups. However, adapting this possibility, there arise three questions: (i) why are beginner learners sensitive to manner/result complementarity, (ii) why are beginner learners insensitive to the generalization (32), and (iii) how do they come to know the generalization?

The immediate answer to the first question (i) is to attribute the sensitivity to L1 transfer; that is, as the dichotomy of manner and result verbs is a universal property, the interlanguage grammar of L2 learners assumes that the dichotomy holds even in L2. This transfer analysis gives another question of why manner/result complementarity is transferred from L1, while the generalization (32) is not. If the generalization (32) is also universal for all languages, it is rather puzzling that the generalization does not transfer from L1 to L2. If the generalization is not transferred, it is reasonable to assume that the knowledge of the manner/result distinction is also not transferred.

Then, how can we give a uniform answer to the questions (i–iii)? I suggest that manner/result complementarity is a part of UG so that the manner/result distinction exists in the learner's interlanguage from the beginning. In this case, what learners must acquire is the generalization (32).¹⁵ In other words, since the generalization is not a part of UG, it must be acquired through linguistic experience. This can be an answer to the question (ii). When it comes to (iii), learners can reach the generalization by repeated exposure to the result verbs appearing in both causative and anticausative constructions and lack of exposure to the manner verbs alternating. Given that learners rarely produce

¹⁵ As Kawaguchi Kenta (p.c.) correctly points out, logically learners can reach a generalization such as (i).

- (i) Manner verbs cannot enter the causative alternation.

This possibility is difficult, if not impossible, to maintain. In order for L2 learners (and L1 learners as well) to reach the generalization (i), learners require negative evidence, given that positive evidence is useless; that is, they need to be explicitly taught that manner verbs do not alternate (i.e., direct negative evidence). However, it is highly unlikely that learners are taught that verbs are divided into manner and result verbs in the first place and that manner verbs do not alternate.

over(anti)causativization errors with manner verbs (see also Section 3), it is clear that they do not wrongly assume that manner verbs can alternate.

6.2 U-Shape Development

Summarizing so far, L2 English learners show different behavior in terms of non-target-like anticausative sentences depending on their proficiency: those at the early stage similarly judge both transitives and causatives in the anticausative frame, whereas those at the next stage reject transitive ones more successfully than causative ones. The interesting developmental difference reminds us of the Unaccusative Trap Hypothesis (UTH), advocated by Oshita (1997, 2001). This subsection briefly reviews the developmental trajectory of unaccusativity in L2 acquisition and suggests that the results of this study show a similar developmental pattern.

6.2.1 The Unaccusative Trap Hypothesis and U-Shaped Development

U-shaped development is one of the very intriguing phenomena observed in both L1 and L2 acquisition. When learners who did not make any errors at the early stage of language acquisition begin to make errors at the middle stage and recover from these errors, U-shape development is observed. This mysterious development is so rare that few cases of U-shape have been reported. In L2 acquisition, the acquisition of unaccusativity is expected to show a U-shape. The U-shape is predicted by the Unaccusative Trap Hypothesis (UTH), proposed by Oshita (1997, 2001) to account for various phenomena found in the L2 acquisition of unaccusativity. According to the UTH, L2 learners undergo three developmental paths: at the initial stage, they fail to distinguish between unaccusatives and unergatives at the level of argument structure, at the second stage, they become aware of the distinction and start making errors related to unaccusatives, such as overpassivization (e.g., **The accident was happened.*), and at the final stage, they recover from these errors.

Throughout the three stages, a U-shape pattern in the development of L2 acquisition of unaccusativity is expected to emerge with respect to passivization errors (Oshita 1997, 2001, 2014, 2022). Learners do not make errors with unaccusatives at the

first stage, start to make passive errors at the second stage, and recover from the errors at the final stage (for L2 English research on overpassivization, see Ju 2000; Oshita 2000; Zobl 1989). At the second stage, passivization errors are expected to occur exclusively with unaccusative verbs but rarely with unergative verbs. This is because overpassivization is argued to occur due to the structural similarity between passives and unaccusatives. In a generative framework, the surface subject of passives and unaccusatives is analyzed to be a derived subject, meaning that the subject is originally generated in the verb's complement (Burzio 1986; Levin & Rappaport Hovav 1995). L2 English learners strongly associate overt marking of passive sentences with NP-movement, overgeneralizing the passive morphosyntax *be + en* to NP-movement in unaccusative sentences.

However, contrary to the prediction of Oshita, previous studies attempting to find U-shaped development failed to observe the pattern (e.g., Deguchi & Oshita 2004; Park & Lakshmanan 2007; Yamakawa et al. 2003). Based on these findings, Oshita (2014) divided a U-shape into an absolute and a relative U-shape, noting that the UTH does not predict an absolute U-shaped development, but rather a relative U-shape development. In his definition of an absolute U-shape, more passive errors with unaccusatives are found in the middle stage of development, compared with those in the initial and final stages. On the other hand, in the definition of a relative U-shape, overpassivized unaccusatives appear more frequently than overpassivized unergatives in the middle, but the differences in passive error rates between unaccusatives and unergatives are not observed in the initial and final stages. Under the notion of a relative U-shaped curve, learners at the middle stage do not become worse at judging passive unaccusatives than when they are at the initial stage, but their judgment of overpassivized unaccusatives deteriorates compared to overpassivized unergatives. The UTH's prediction was confirmed in Oshita's (2014) cross-sectional study. He observed a clear U-shaped development in which L2 English learners judged passive unaccusatives less correctly than passive unergatives at the middle proficiency level.

6.2.2 U-Shape in the Acquisition of the Causative Alternation?

The current study showed a similar developmental tendency to unaccusativity. In L2 English, overpassivized unaccusatives are similarly rated to overpassivized unergatives at the initial stage, and the overpassivized unaccusatives begin to be judged as more acceptable compared with overpassivized unergatives at the next stage. In this study, the low proficiency group showed a similar judgment of overanticausativized causatives and overanticausativized transitives (see Figure 2), while the intermediate proficiency group rated overanticausativized causatives more acceptable than overanticausativized transitives (see Figure 3).

To see if the difference in acceptability between the difference between overanticausativized causatives and transitives in the Low group and the difference between overanticausativized causatives and transitives in the Intermediate group is statistically significant, one-way ANOVA was conducted on the four experimental groups, using *anovakun* (version 4.8.9). Following Oshita (2014), the mean z-scores of overanticausativized transitives were subtracted from the mean z-scores of overanticausativized causatives before the statistical analysis, so that how differently participants treated the two types of verbs can be analyzed.¹⁶ Before conducting the ANOVA, the assumptions of homogeneity of variances were tested. Levene's test showed homogeneity of variances ($F(3, 81) = 1.21, p = .31$). Therefore, a one-way ANOVA was conducted. The results of one-way ANOVA did not reveal any significance ($F(3, 81) = 0.368, p = .777, \eta^2 = 0.013$). Figure 8 shows a U-shape development, though not so clear, but it does not receive support from statistical analysis.

¹⁶ We do not focus on an absolute U-shape here because this type of U was not observed, as seen in Figure 7. However, numerically the High group was the worst in rejecting anticausative sentences. It might be a possibility that L2 English learners start to make anticausativization errors at the relatively late stage of development. I leave this question for the future study.

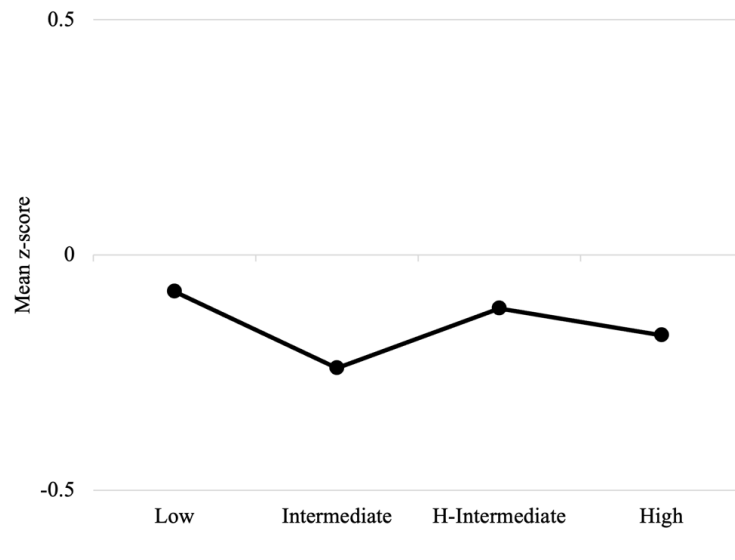


Figure 8. Group Differences in Judgment of Anticausative Sentences

Summarizing, a relative U-shape (as well as an absolute U) was not observed in our study (at least not statistically supported), although the intermediate proficiency group showed a less accurate performance on ungrammatically anticausativized causative verbs. The different behavior of the intermediate proficiency learners, however, is interesting, and further investigation of U-shape development is called for.

Chapter 7: Conclusion

In the previous research, causative alternation errors—both overcausativization and overanticausativization errors—have long been studied. Causativization errors, such as **Brian fell the cup from the table*, have been reported on many occasions, while anticausativization errors, such as **The building destroyed*, have received comparatively less attention in the literature. To account for these errors, two analyses are often adapted. One attributes these errors to L1 transfer of lexical entries (see Section 4.1.1), and the other to non-target assignment of lexical-semantic structure (see Section 4.1.2). In this thesis, I also proposed a novel analysis, building on the manner/result complementarity (see Section 4.2). The analysis assumes that L2 learners are sensitive to the semantic differences between manner and result verbs due to UG, and that they also know that only result verbs can enter the causative alternation. Based on these assumptions, it posits that causative alternation errors occur because of the cross-linguistic differences in which result verbs alternate. Depending on the language, alternating result verbs differ; simply put, a given verb that alternates in one language does not alternate in the other. Because of the differences, learners must acquire which verbs alternate from experience. However, since they know that only result verbs are qualified for alternation, they incorrectly produce and accept over(anti)causativized result verbs.

The important prediction of this analysis is that learners predominantly make such errors with result verbs, but fewer with manner verbs. To test this prediction, I conducted an acceptability judgment task with 120 Japanese-speaking learners of L2 English. The results showed that intermediate proficiency learners were more accurate at rejecting overanticausativized manner transitive verbs than overanticausativized result transitive verbs, supporting the novel analysis (see Section 5).

The results also suggested that learners appear to undergo a U-shaped development: they initially judge overanticausativized result and manner verbs similarly, but at the next stage, they start to be more accurate at rejecting overanticausativized manner verbs than overanticausativized result verbs. This U-shape pattern, however, did not receive statistical support (see Section 6.2), requiring further investigation for it.

Overall, the results of this study (especially from the intermediate proficiency learners) suggest that L2 learners are sensitive to the manner/result distinction and, at some point, reach the generalization that only result verbs can alternate. Learners' sensitivity to the manner/result is puzzling, given that the existence of two types is rarely taught in a classroom environment, posing a logical problem of acquisition (see White 2003). To explain the problem, I posit that manner/result complementarity is a part of UG.

Before concluding, I give several limitations of this study. For one, this study conducted only a judgment task without supplementary tasks, such as correction, in which participants correct sentences that they judged as incorrect. One primary reason why I did not incorporate these correction tasks was to reduce participants' burden and shorten the total length of the task (see also Section 5.5). However, due to the lack of a correction task, we cannot ensure that participants judged sentences based on whether a given verb can be used in the (anti)causative construction; that is, they potentially judged a given sentence as (in)correct by looking at the use of, say, articles. This methodological limitation, however, seems to me not so problematic, given that the results showed that all the proficiency groups judged grammatical transitive/causative sentences as correct and ungrammatical anticausative sentences as incorrect.

Another limitation concerns the frequency of experimental verbs. (Anti)causativization errors have been claimed to be affected by the frequency of the verb (Nagano, 2015; Zyzik & Bittar, 2024): the more frequent a given verb is in the input, the more accurate L2 learners' judgment is. Due to the limited number of English non-alternating result transitive verbs that are also non-alternating in Japanese, the current study could not take the frequency factor into account.

Finally, this study assumes that over(anti)causativization errors are not limited to a specific L1 group but are universal errors committed by any L1 speakers. To ensure the assumption, it would have been better to include not just L1 Japanese-speakers but also other L1 groups to compare. These limitations listed here should be addressed in future studies in order to provide a comprehensive understanding of the acquisition of the causative alternation.

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Appendix

List of experimental sentences with contextual paragraphs (experimental sentences were italicized)

Result verb:

1. The elephant inside the strong yellow fence at the zoo suddenly went wild. The elephant rushed at the fence and...

The angry elephant destroyed the yellow fence.

A strong yellow fence was built around the elephant in the zoo. However...

The yellow fence destroyed last night.

2. Last month, a new bridge was built over the river in Nagoya. Then a huge monster appeared. And then...

The huge monster destroyed the new bridge.

A new bridge was built over the river in Nagoya. However...

The new bridge destroyed last night.

3. The young chef found spoiled milk in the refrigerator of the restaurant. And then...

The young chef removed the spoiled milk.

Since last week, spoiled milk had been in the refrigerator of the Italian restaurant.

However...

The spoiled milk removed today.

4. The old librarian found a very heavy book in the library. And then...

The old librarian removed the heavy book.

For 20 years, a very heavy book had been placed on the library shelf. However...

The heavy book removed today.

5. John took his energetic female cat to his friend' s house. In that house, there was an expensive curtain. And then...

The female cat damaged the expensive curtain.

John hung a beautiful curtain in his own room. However, for some reason...

The expensive curtain damaged in the morning.

6. Brian took his energetic male dog to his friend' s house. On the sofa at his friend' s house, there was a beautiful dress. And then...

The male dog damaged the beautiful dress.

Yesterday, Mary bought a beautiful dress. However, for some reason...

The beautiful dress damaged in the morning.

7. The careful construction worker was asked to build a brick wall. The next day...

The careful builder created the brick wall.

It was planned that a brick wall would be built in the university parking lot. And then...

The brick wall created two days ago.

8. The skilled engineer was asked to make a robotic arm. And then...

The skilled engineer created the robotic arm.

Three days ago, parts to make a robotic arm were placed on the factory table. And then...

The robotic arm created two days ago.

9. The talented architect was asked to build a modern tower. And then...

The talented architect constructed the modern tower.

The construction of a modern tower had been planned. And then...

The modern tower constructed yesterday.

10. A creative student drew blueprints for a metal building. And then...

The creative student constructed the metal building.

The construction of a metal building had been scheduled. And then...

The metal building constructed yesterday.

Manner verb:

1. A curious baby found a round mirror in the corner of the room. And then...

The curious baby touched the round mirror.

Three days ago, in the center of a tourist spot, a fine, round mirror that must not be touched was placed. However...

The round mirror touched three days ago.

2. A curious child found a magic box in front of him. And then...

The curious child touched the magic box.

Three days ago, in an alley, a magic box that must not be touched was placed. However...

The magic box touched three days ago.

3. The famous scientist bought a special camera to observe a certain phenomenon. And then...

The famous scientist used the special camera.

In the university laboratory, there was a special camera that no one was allowed to use. However...

The special camera used yesterday.

4. The famous scientist bought a dangerous machine for his research. And then...

The famous scientist used the dangerous machine.

In the university laboratory, there was a dangerous machine that no one was allowed to use. However...

The dangerous machine used yesterday.

5. A lazy soldier stood in front of a red button. And then...

The lazy soldier pushed the red button.

At the office, there was a red button that was said must never be pressed.

However...

The red button pushed in the morning.

6. A strong boy stood in front of a small rock that would never move no matter what.

And then...

The strong boy pushed the small rock.

In Canada, there was a small rock that was said must never be pushed. However...

The small rock pushed in the morning.

7. An employee managing a valuable sculpture in the museum was suddenly told he was fired, and he got angry. And then...

The fired employee kicked the valuable sculpture.

In the museum, a valuable sculpture had been protected so that no one could damage it. However...

The valuable sculpture kicked last Sunday.

8. An excited tourist was acting violently around an ancient statue in London. And then...

The excited tourist kicked the ancient statue.

In London, there was an ancient statue that had been protected for a long time so that no one could damage it. However...

The ancient statue kicked last Sunday.

9. A rich journalist test-drove a white car. And then...

The rich journalist purchased the white car.

A white car displayed at the dealership had not been sold at all for five years. However...

The white car purchased today.

10. A Japanese singer test-drove a popular motorcycle. And then...

The Japanese singer purchased the popular motorcycle.

A motorcycle that was popular five years ago had not sold at all recently. However...

The popular motorcycle purchased today.